

# Monitoring forest disturbance using CCDC-SMA on Google Earth Engine

## Main objective:

The goal of this lab is to use CCDC-SMA to monitor forest disturbance. CCDC-SMA stands for Continuous Change Detection and Classification - Spectral Mixture Analysis. It used fractions of endmembers (GV, NPV, Soil and Shade) and NDFI as the inputs for the CCDC algorithm. For more information on CCDC-SMA, please read the paper and websites provided on Blackboard.

In this lab, you will:

- Collect forest and non-forest training data on GEE for the start year of your study period.
- Use the training data to make a forest mask.
- Run CCDC-SMA using the forest mask you created.
- Visually assess the results of CCDC-SMA and compare it with the results of LandTrendr from previous lab

## Getting ready:

You should be quite familiar with the GEE interface by now. Follow these steps to set up your workspace for this lab:

- Click the link below to add the GEE repository to your reader repositories (Under "Script" tab, in the "Reader" list).
  https://code.earthengine.google.com/?accept_repo=users/shijuanchen32/share
- In the "DIP_labs" folder, there are two scripts for this lab: *Lab11_1_CCDC-SMA_prep, Lab11_2_CCDC-SMA_compare*. Copy and paste the two scripts into your own repository.
- Read the instructions of how to run CCDC-SMA apps on the GitHub page below, and click the link of the repository of CCDC-SMA on that page to add it in your reader repository. GitHub page link:
  https://github.com/shijuanchen/forest_degradation_georgia

## Define and export your study region:

Click your copy of the script of *Lab11_1_CCDC-SMA_prep*, draw a study region, named "region". You need to specify the asset id to export. If you've ready have a study region from the previous lab (LandTrendr), I'd recommend you to use the same one.

## Create a forest mask of the start year:

To create a forest mask, you need to collect training data of forest and non-forest of the start year of your study period. Click your copy of the script of *Lab11_1_CCDC-SMA_prep*. Create a layer named *forest* and another layer named *non_forest*. Choose "FeatureCollection" as the type. In the properties, give a property name and value (Fig. 1.).

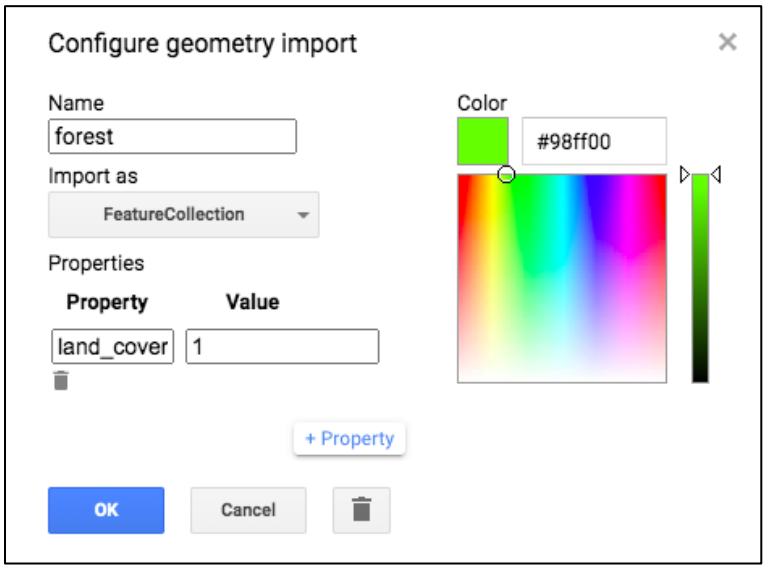



Fig. 1 Configure property of training data

Customize your inputs in line 17 - line 25:

```
/* --------------------Customize your inputs---------------------------------*/
var region_id = 'users/shijuanchen32/DIP/lab11_region';
var start_year = 1990;
var property = 'land_cover';
var training = forest.merge(non_forest);
var forest_value = 1;
var non_forest_value = 2;
var forest_id = 'users/shijuanchen32/DIP/lab11_forest';
/* --------------------------------------------------------------------------*/
```

Run the script to add an image composite using median surface reflectance of the start year as a map layer. You may get error message in the "Console" because the training data is empty. Don't worry about it for now, and let's look at the image composite and start to collect the training data (Fig. 2). After you collect some training data, run the script again, you won't get any error. A classification map will be added to the map layers. In the task tab, click "run" button to save your study area and forest mask as an GEE asset. Now, you are ready to run CCDC-SMA.

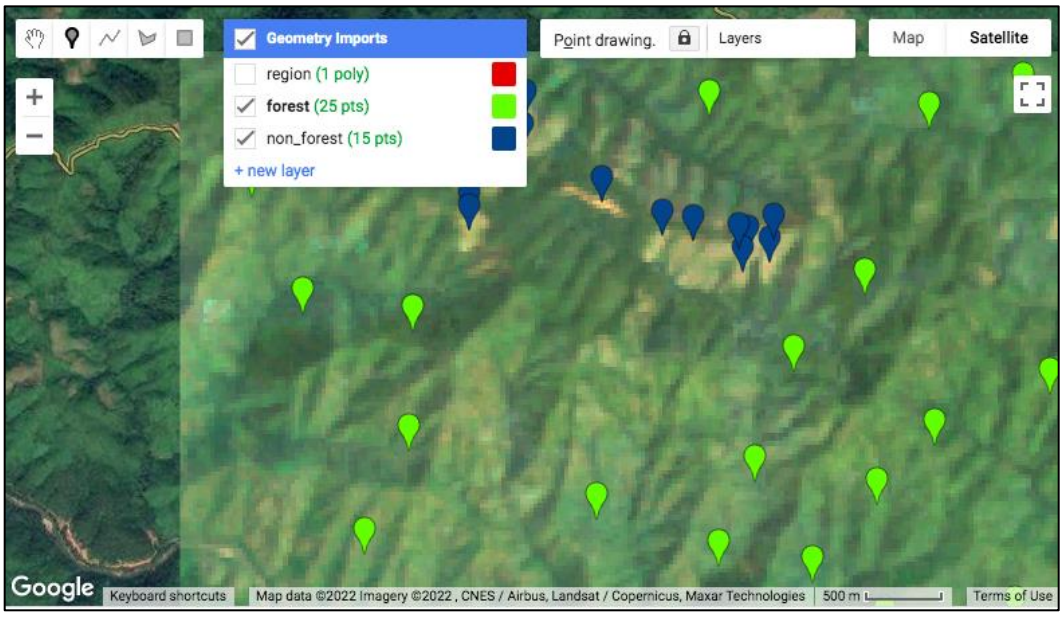



Fig. 2 Collect training data of forest and non-forest

## Run CCDC-SMA:

You will choose one version of CCDC-SMA apps. Enter the GEE asset id of your study region and the forest mask. Define the start and end year of your study period. For the output, you can either only export the first year of disturbance or both the first year and the full segment results. For this lab, exporting the first year of disturbance is enough. Click "Run CCDC-SMA", and submit the task(s). After your task is finished, you will get GEE assets of the results. Enter the asset id of your results in the "View results" section of the CCDC-SMA app, you will be able to see your results.

## Visually assess CCDC-SMA and compare it with LandTrendr:

In your copy of the script named *Lab11_2_CCDC-SMA_compare*, enter your own asset id of CCDC-SMA results of the first year of disturbance, and the asset id of the LandTrendr results from the previous lab. Enter your start and end year, and the asset id of your study region. Run the script, visually compare these two results in the forest regions. The two image composites might help you to assess your results, but note that if vegetation recovers after a disturbance, you may not be able to see this disturbance by comparing the two image composites. Also note that here we apply a forest mask for CCDC-SMA, but did not use a forest mask with LandTrendr in the previous lab.